

Seeing Together: Multi-Robot Cooperative Egocentric Spatial Reasoning with Multimodal Large Language Models

Kunyu Peng^{1,2,*} Zhikun Zhou³ Kailun Yang^{3,*} Di Wen¹ Ruiping Liu¹
Yufan Chen^{1,4} Junwei Zheng^{1,6} Hao Shi^{5,7} Yi Zhou³
M. Saquib Sarfraz¹ Danda Pani Paudel² Luc Van Gool²

¹Karlsruhe Institute of Technology ²INSAIT, Sofia University “St. Kliment Ohridski”

³Hunan University ⁴University of Oxford ⁵Zhejiang University ⁶ETH Zurich ⁷Ant Group

Abstract

Multimodal Large Language Models (MLLMs) have made substantial progress in egocentric video understanding, but their ability to reason cooperatively from multiple embodied viewpoints remains largely unexplored. We study this problem through *multi-robot cooperative dynamic spatial reasoning*, where a model must answer spatial, temporal, visibility, and coordination questions by integrating synchronized egocentric videos from a team of moving robots. To support this setting, we introduce **CoopSR**, the first benchmark for this task, together with **EgoTeam**, a multi-robot egocentric QA dataset. EgoTeam contains 114,227 QA pairs spanning 19 question types, four difficulty tiers, and three team sizes in Habitat and iGibson, along with a real-world test set of around 2,326 QAs collected using two quadruped robots. We further propose **SP-CoR** (*Spectral and Physics-Informed Cooperative Reasoner*), an MLLM framework for fine-grained cooperative spatial reasoning. SP-CoR combines dynamics-aware multi-robot frame sampling, spectral- and physics-guided view fusion, and physics-aligned prompt distillation, enabling the model to benefit from privileged robot-pose supervision during training while requiring only egocentric videos at test time. Across 22 MLLM baselines, SP-CoR consistently improves cooperative reasoning, outperforming the strongest fine-tuned baseline by **+3.87%** on Habitat and **+7.12%** on iGibson. It also shows stronger generalization to unseen team sizes and real-world robot tests. Code can be found at <https://github.com/KPeng9510/seeing-together.git>.

1 Introduction

Embodied AI is rapidly shifting from isolated agents to teams of robots operating in shared physical spaces, including warehouse fleets, search-and-rescue squads, inspection drones, and household assistants [63, 18, 10, 29]. In these settings, no single robot has a complete view of the world. Each agent observes only a partial, egocentric, and often noisy slice of the scene. Answering even a seemingly simple operator query, such as “Which robot is pushing the object?” or “Can any robot see the blocked doorway?”, may require integrating observations across robots, grounding objects across viewpoints, reasoning about inter-robot spatial relations, and building a team-level understanding of the scene. Beyond perception alone, real-world deployment also requires robots to coordinate tightly during physically coupled tasks, such as two robots jointly carrying a long object through cluttered spaces, where failures in shared spatial understanding can lead to collisions, blockage, or unsafe interactions. We call this capability *cooperative spatial reasoning*: reasoning over multi-robot egocentric video streams to build a shared understanding of a dynamic environment.

*Correspondence: kunyu.peng@kit.edu and kailun.yang@hnu.edu.cn

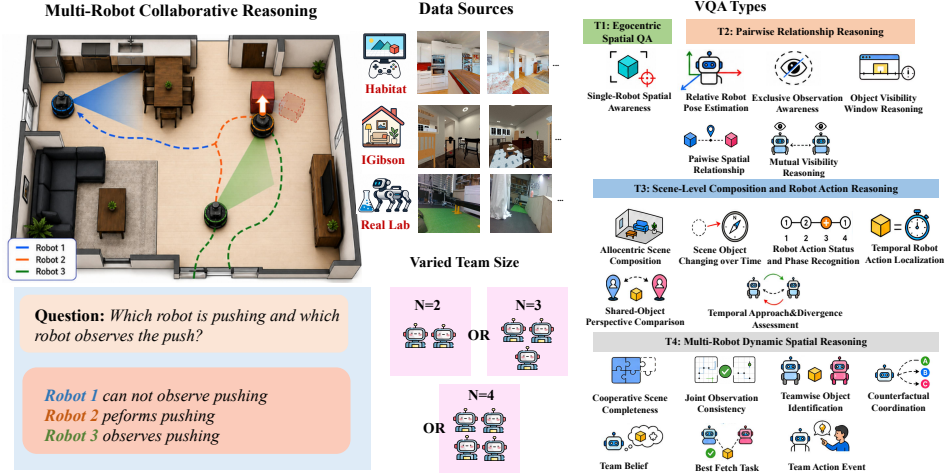


Figure 1: Overview of the **CoopSR** benchmark. It evaluates spatial, temporal, visibility, and coordination reasoning using synchronized egocentric views from variable-size robot teams in simulated and real environments.

Despite its importance, this setting remains largely unexplored in current Multimodal Large Language Models (MLLMs). Existing MLLMs [68, 63, 72, 24] are primarily trained and evaluated on single-view images or videos, often from a single human or robot perspective. They typically treat visual tokens as a flat sequence and lack mechanisms for explicitly aligning, filtering, and fusing observations from multiple embodied agents. A naive extension—simply concatenating N robot video streams—is not scalable: the number of visual tokens grows rapidly with team size, while adjacent frames and overlapping viewpoints introduce substantial redundancy. More importantly, concatenation alone does not teach the model *how* to reason collaboratively across agents, such as determining which robot has the best viewpoint, resolving contradictory observations, or inferring scene-level structure from distributed egocentric evidence.

Existing benchmarks [16, 27, 50] do not adequately address this problem. Egocentric VQA benchmarks [50, 12] mainly consider a single observer, while EgoExo QA benchmarks [28, 21] focus on reasoning between a fixed exocentric view and one moving egocentric view. Recent VQA benchmarks for moving egocentric agents, *e.g.*, MA-EgoQA [29], assume that the agent already possesses sufficient spatial reasoning ability in dynamic scenes and primarily emphasize lifelong human behavior understanding, rather than fine-grained joint spatial reasoning, robot action understanding, and environment awareness. In contrast, our benchmark is uniquely designed to evaluate whether MLLMs can understand a shared dynamic environment through the eyes of a robot team. It concentrates on cross-view grounding, inter-robot spatial relations, team-level scene understanding, and fills a critical gap by directly assessing team-level egocentric perception and reasoning for multi-robot embodied intelligence.

To fill this gap, we introduce **CoopSR**, a benchmark for *cooperative multi-robot visual question answering*, and **EgoTeam** dataset collected from both simulators and real-world labs. **EgoTeam** contains 114K QA pairs covering 19 QA types along five axes: spatial, exploration, relational, robot action, and visibility reasoning. The questions span four levels, from egocentric scene understanding to cooperative scene perception, and three team sizes in habitat, iGibson, and real-world environments. **CoopSR** evaluates 22 MLLMs on synchronized egocentric observations from variable-size robot teams, targeting key skills such as spatial awareness, robot-object relations, scene composition, temporal robot action reasoning, mutual visibility, shared-object perception, team belief updates, and counterfactual coordination. Unlike prior VQA benchmarks focused on single-agent perception, **CoopSR** evaluates whether models can integrate partial multi-robot viewpoints, reason about complementary observations and inter-robot relations, and infer coherent scene-level answers from distributed embodied evidence.

However, existing models [26, 68, 72, 24, 1, 59] perform reasonably well on single-view reasoning [12, 50] or EgoExo setting [23] but degrade sharply on questions requiring inter-robot integration, cross-team-size generalization, or conflicting viewpoints. This failure pattern suggests a structural bottleneck: current MLLMs lack inductive biases for agent identity, geometric co-visibility, and

multi-robot egocentric view fusion. Motivated by these findings, we propose **SP-CoR** (*Spectral and Physics-Informed Cooperative Reasoner*), an MLLM architecture designed for cooperative spatial reasoning. **SP-CoR** has three key components. First, a training-free two-stage sampler compresses each robot stream by jointly selecting the most query-relevant frames and refining them using temporal frequency energy ranking over visual features. Second, a spectral and physics-informed attentional block fuses robot observations while injecting a physics prior from the data-rich simulator and the spectral cues into attentional fusion. Ground-truth poses are used only during training, and deployment requires no external localization oracle. Third, a pose-aware teacher distills physical priors into the student’s prefix tokens, aligning the language prompt with the pose prior provided by the simulator. The student prompt is utilized as physics prior during deployment, as the real world cannot be guaranteed to provide precise pose cues.

On the **CoopSR** benchmark, **SP-CoR** consistently outperforms zero-shot, SFT, RAG [29, 26], and key frame selection [56, 57] baselines across different evaluation settings. The best **SP-CoR** variant, using a Qwen2.5-VL [2] backbone, achieves an average accuracy of 70.55% on Habitat [48] and 70.82% on iGibson [32], improving over the strongest baseline by +3.87% and +7.12%, respectively. **SP-CoR** also shows superior cross-MLLM-backbone generalizability. These results indicate that **SP-CoR** benefits from an explicit cooperative reasoning design. Further evaluation on cross-team-size and sim-to-real generalizability further demonstrates the superiority of our proposed method. In summary, our contributions are as follows:

- We introduce **cooperative spatial reasoning** as a new embodied AI problem, where an MLLM must answer language queries by integrating synchronized egocentric observations from multiple moving robots.
- We present **EgoTeam**, a dataset for cooperative multi-robot visual question answering, containing > 114K question-answer pairs across 19 QA types in simulated and real environment. We conduct a comprehensive evaluation of 22 MLLM baselines on this dataset and build this first **CoopSR** benchmark.
- We propose **SP-CoR**, a cooperative MLLM framework that introduces a spectral energy-aware multi-robot relevant frame sampler, spectral and physics-aligned multi-robot fusion, and physics-aligned prompt-space distillation for scalable reasoning over multi-robot egocentric videos without requiring pose data during test time. **SP-CoR** delivers state-of-the-art performance across Habitat, iGibson, cross-team-size, and sim-to-real settings.

2 Related Work

Multi-agent perception and cooperation. Existing works on multiple embodied agents study cooperation across diverse environments [39, 62, 73, 6, 49, 71, 53], with systems such as CoELA [70], Co-NavGPT [65], OmniVLN [38], DeCoNav [74], AirCopBench [67], TZPP [61], COHERENT [36], and CoMaTrack [37] using LLMs/VLMs for multi-agent planning, coordination, and exploration. Other studies address more realistic constraints such as limited communication and partial observability [66, 25], while structured or leadership-based prompting has been shown to improve teamwork efficiency [20]. PARTNR [8] further provides a large-scale human-robot collaboration benchmark.

Multimodal large language models. The past two years have seen rapid progress in MLLMs that combine vision encoders with LLM backbones: Video-LLaVA [33], InternVL [11], Qwen2.5-VL [2], Qwen3-VL [1], LLaVA-Next-Video [72], and VideoLLaMA3 [68]. These models are trained on single-image or short-video data and treat all visual tokens uniformly, with no mechanism for agent-level identity or inter-agent geometric constraints. We systematically evaluated 22 MLLMs, including zero-shot MLLMs, supervised fine-tuned MLLMs, Retrieval-Augmented-Generation methods [29, 26], and key frame selection methods [57, 56]. Existing MLLMs lack cross-robot alignment, multi-view fusion, and team-level belief modeling, therefore, we propose **SP-CoR**, which integrates dynamics-aware sampling, geometry-guided fusion, and physics-aligned prompt distillation for cooperative spatial reasoning.

Benchmarks of egocentric video question answering. Early egocentric vision research [15, 17, 30, 52, 3, 43, 45, 35] was driven by foundational but relatively small datasets such as ADL [46]. The field later expanded through large-scale benchmarks including EPIC-KITCHENS [13], Ego4D [18], and EPFL-Smart-Kitchen [5], which broadened egocentric video understanding to diverse daily activities. Subsequent datasets targeted more specialized settings, including procedure learning [4, 54, 75],

Table 1: Comparison of our **EgoTeam** with other egocentric QA datasets.

Dataset	Source	#QAs	Size (hrs)	#Clips	Dur./Clip	MultiView	Inter Robot Dynamic	Robot Pose
EgoSchema [42]	Ego4D [18]	5K	250	5,063	3 min	✗	✗	✗
EgoPlan-Bench [10]	Ego4D [18] & EpicKitchen [13]	4.9K	-	4,939	-	✗	✗	✗
EgoThink [12]	Ego4D [18]	0.7K	-	595	-	✗	✗	✗
EgoMemoria [64]	Ego4D [18]	7K	-	629	30 s to 1 h	✗	✗	✗
HourVideo [7]	Ego4D [18]	12.9K	381	500	20 min to 2 h	✗	✗	✗
EgoLifeQA [63]	EgoLife [63]	3K	266	6	44.3 h	✓	✗	✗
EgoTeam (ours)	-	114K	385.7	21,124	1 min to 6 min	✓	✓	✓

collaborative assistance [60], and joint egocentric-exocentric understanding [19, 23]. Building on these resources, recent benchmarks have advanced temporal reasoning, forecasting, and planning in first-person video [34, 14, 47, 58, 42, 31]. Zhang *et al.* [69] propose pairwise egocentric video question answering in day and night time. More recent efforts further explore long-term video understanding [64, 7]. Jung *et al.* [28] and He *et al.* [21] focus on egocentric and exocentric videos for cross-view video understanding. MA-EgoQA [29] focuses on long-term human behavior reasoning using multiple agents, which focuses on coarse-level QAs. Unlike these benchmarks, our **CoopSR** directly targets fine-grained cooperative spatial reasoning from synchronized multi-robot egocentric videos. It evaluates whether MLLMs can integrate partial and dynamic robot viewpoints to reason about inter-robot relations, shared scene structure, visibility, and team-level coordination.

3 CoopSR-Bench: Benchmark and Datasets

Data collection and annotation. Metadata collection. We first collect multi-robot egocentric data in Habitat [48] and iGibson [32]. For each scene, we deploy teams of 2, 3, and 4 robots and generate 15 exploration episodes with diverse trajectories and robot-object interaction events, such as object pushing, yielding diverse layouts, viewpoints, and temporal dynamics. For each episode, we record robot poses, pairwise relative poses, egocentric RGB-D videos, semantic segmentation videos, and object to object/robot spatial relationships per timestep. These annotations support QA construction over egocentric perception, cross-view association, temporal reasoning, and multi-robot coordination. We also collect a real-world test set by exploring a room with two quadruped robots and using motion capture systems to provide pose metadata for QA creation; details are provided in the appendix.

QA annotations. For iGibson [32] and Habitat [48], we use simulator-recorded scene attributes as metadata for QA generation and then rephrase the QAs by GPT-4o [44]. QA pairs are derived from the aforementioned extracted metadata. This enables scalable QA generation while ensuring that each question is grounded in the scene state and has an unambiguous metadata-derived answer. For the real-world quadruped robot dataset, we manually annotate scene captions as metadata at adaptive intervals of fewer than 30 frames to avoid obvious scene changes within each interval, and then create QAs based on metadata. In total, we collect 1.75h of real-world data across 13 scene settings. Additional statistics and comparisons with existing egocentric datasets are shown in Table 1. To improve evaluation reliability, four annotators manually review QA pairs in the validation and test splits to ensure correctness with a majority agreement ratio of 0.9.

QA taxonomy. Our QA taxonomy evaluates multi-robot egocentric reasoning across four progressive levels, ranging from local spatial perception to team-level coordination, as illustrated in Figure 1.

T1: Egocentric spatial QA. T1 level evaluates spatial understanding from a single robot’s first-person observation. It includes single-robot spatial awareness, such as recognizing object locations, directions, distances, and layouts.

T2: Pairwise relationship Reasoning. T2 level focuses on relationships between two entities, including robots, objects, and viewpoints. It evaluates whether a model can identify information visible to only one robot, reason about object visibility and occlusion over time, determine pairwise spatial relations such as left/right, front/behind, near/far, and facing direction, and assess whether two robots can see each other or share overlapping fields of view.

T3: Scene-level composition and robot action reasoning. T3 level requires integrating multiple egocentric observations into a coherent scene-level representation. It covers allocentric scene reconstruction, temporal tracking of object changes, recognition of robot action phases, estimation of action duration, comparison of shared objects across viewpoints, and reasoning about whether robots approach or move away from one another or from target objects over time.

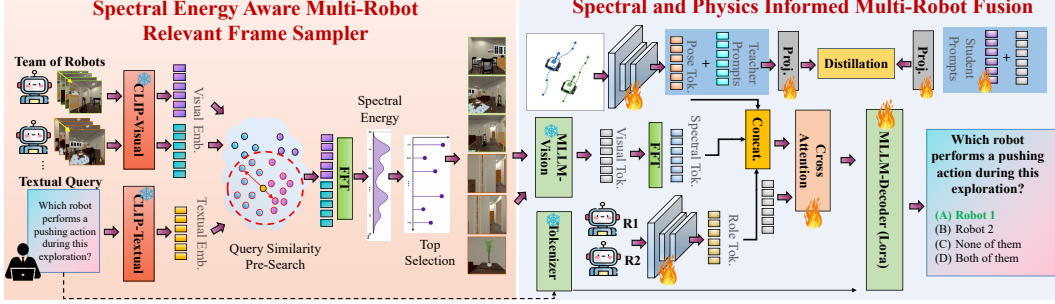


Figure 2: Overview of **SP-CoR** for **CoopSR**: a query-guided spectral energy sampler selects informative multi-robot egocentric frames, while spectral- and physics-informed fusion with prompt distillation integrates robot-view evidence for cooperative spatial reasoning.

T4: Multi-robot dynamic spatial reasoning. T4 level evaluates high-level collaborative reasoning over the full robot team. It tests whether the model can assess collective scene coverage, detect consistency or conflict among robot observations, identify the same object across views, reason counterfactually about alternative coordination strategies, update team-level beliefs from new observations, assign fetch tasks based on visibility, distance, and action context, and infer the overall team event, including acting and observing robots. Overall, the taxonomy provides a structured evaluation of multi-robot egocentric reasoning, progressing from single-view perception to pairwise relations, global scene understanding, and dynamic team coordination.

Baselines. Zero-shot MLLMs: We evaluate representative image/video MLLMs, including Qwen2.5-VL-7B/72B [2], Qwen3-VL-8B [1], InternVL3-8B [59], LLaVA-NeXT-Video-7B [72], and VideoLLaMA3-7B/2B [68], to assess their zero-shot performance on **CoopSR**. Supervised fine-tuning baselines: We fine-tune the same major backbones with Low Rank Adaptation (LoRA) [22] on the MLLM decoder on **CoopSR** to test whether standard task adaptation is sufficient and to isolate the architectural gains of **SP-CoR**. Retrieval-augmented baselines: We compare with VideoRAG [41] and EgoRAG [63] to evaluate whether retrieved video or egocentric evidence improves reasoning over long, redundant multi-robot streams. Keyframe-selection baselines: We include TSPO [57] and KTV [56] under the best model from Supervised Fine-Tuning (SFT) to examine the performance of keyframe selection approaches. Together, these baselines control for *model capability*, *task adaptation*, and *evidence selection*, allowing us to verify that **SP-CoR** improves **CoopSR** through cooperative inductive biases for agent identity, co-visibility, temporal dynamics, and multi-robot fusion.

Dataset splits and evaluation metric. Data splits. For the dataset with 114, 227 QAs collected from Habitat [48], we use 119/12/23 unique scenes for train/val/test, while the iGibson [32] counterpart use 8/2/4 scenes for train/val/test. All 13 scenes of the real-world set are used as the test set, which contributes to 2, 326 QAs in total. Multiple-choice QA (MC4). For QA types 01–19, models produce a single letter (A/B/C/D) and are scored. Primary metrics are MC4 Accuracy (overall and per task-difficulty/team-size level), including per-difficulty-tier accuracy, per-source accuracy (non-iGibson vs. iGibson), and N -robot generalization (train on $N = [2, 3]$, evaluate on $N = 4$). AVG denotes the averaged QA performance on all samples without grouping.

4 Methodology

4.1 Problem Formulation

We study *multi-robot egocentric cooperative spatial reasoning*, where N robots jointly answer a query from synchronized egocentric video streams. Robot r observes $\mathcal{V}_r = \{\mathbf{x}_{r,t}\}_{t=1}^T$ and receives controls $\mathbf{u}_{r,t}$. During training, we additionally use pose estimates $\hat{\mathbf{u}}_{r,t} = (\hat{\mathbf{p}}_{r,t}, \hat{\mathbf{e}}_{r,t}) \in \text{SE}(3)$ from simulators or motion capture; these poses are unavailable at test time. Given $\mathcal{V} = \{\mathcal{V}_r\}_{r=1}^N$ and query $\mathbf{q}$, the model predicts $\mathbf{y} = f_\theta(\mathcal{V}, \{\hat{\mathbf{u}}_{r,t}\}, \mathbf{q})$ during training, $\mathbf{y} = f_\theta(\mathcal{V}, \mathbf{q})$ during testing. T and N denote number of frames and robots. This setting requires selecting sparse evidence across robots and time, resolving redundant or conflicting viewpoints, and reasoning over shared physical constraints.

4.2 Spectral and Physics-Aware Cooperative Reasoner (SP-CoR)

Current MLLMs struggle with multi-robot cooperative spatial reasoning due to limited cross-robot dynamics modeling and weak use of physical priors. We propose **SP-CoR** (*Spectral and Physics-Aware Cooperative Reasoner*), which combines spectral energy-aware frame sampling with physics-informed view fusion and prompt distillation. It selects informative multi-robot frames, injects motion and spatial priors during training, and enables pose-free cooperative reasoning at inference. The whole pipeline of our approach is shown in Figure 2.

Spectral Energy-aware Multi-Robot Relevant Frame Sampler (SE-MR²FS). Feeding all frames from all robots yields $\mathcal{O}(NT)$ visual tokens for N robots and T frames, which is costly and redundant due to adjacent near-duplicate frames and overlapping robot views. We therefore compress all robot streams to a fixed budget of $K \ll T$ frames using a training-free two-stage sampler: a frozen vision–language model first retrieves query-relevant candidates, which are then refined by temporal frequency energy over visual features. Specifically, each frame is embedded with a frozen CLIP [51] image encoder ϕ_v as $\mathbf{z}_{r,t} = \phi_v(\mathbf{x}_{r,t}) \in \mathbb{R}^{d_c}$, while the query is encoded by the corresponding CLIP text encoder as $\mathbf{q} \in \mathbb{R}^{d_c}$. Both embeddings are ℓ_2 -normalized.

CLIP-based semantic candidate selection. For each robot r , we score every frame by its query similarity according to Eq. 1.

$$\mathbf{s}_{r,t} = \langle \mathbf{z}_{r,t}, \mathbf{q} \rangle \in [-1, 1], \quad (1)$$

and retain the top- M frames as a candidate set as shown in Eq. 2.

$$\mathcal{C}_r = \text{TopM}(\{\mathbf{s}_{r,t}\}_{t=1}^T; M). \quad (2)$$

This step removes frames that are obviously irrelevant to the question before any further step.

Temporal FFT energy on visual features. Many questions depend on understanding temporal dynamics, such as which robot moved first, when an object was placed, or which robot interacted with another object or agent. These changes cannot be captured reliably by static CLIP [51] similarity alone. We therefore compute a per-frame motion descriptor by applying a discrete Fourier transform along the temporal axis of CLIP visual features. For a sliding window $\mathcal{W}(t)$ of length w centered at t , let $\mathbf{Z}_{r,\mathcal{W}(t)} \in \mathbb{R}^{w \times d_c}$ denote the stacked CLIP features after subtraction of the local mean. Fourier transformation is conducted according to Eq. 3.

$$\Phi_{r,t}[k] = \log\left(1 + \frac{1}{d_c} \|\mathcal{F}_w[\mathbf{Z}_{r,\mathcal{W}(t)}]_k\|_2^2\right), \quad k = [1, \dots, B], \quad (3)$$

where $\mathcal{F}_w$ is the length- w rFFT, the zero-frequency component is discarded so that the descriptor responds only to change, and the remaining bins are pooled into B bands. The scalar frequency energy of frame (r, t) is the sum of retained bands according to Eq. 4.

$$\mathbf{e}_{r,t} = \sum_{k=1}^B \Phi_{r,t}[k]. \quad (4)$$

Because the FFT operates on CLIP embeddings rather than pixels, $\mathbf{e}_{r,t}$ captures semantically meaningful motion while remaining insensitive to nuisance pixel-level perturbation. We restrict the FFT energy to the candidate set $\mathcal{C}_r$ and standardize it within that set to get $\tilde{\mathbf{e}}_{r,t}$. The final per-frame score combines frequency energy with an optional semantic-refinement term according to Eq. 5,

$$\rho_{r,t} = \tilde{\mathbf{e}}_{r,t} + \tilde{\mathbf{s}}_{r,t}, \quad t \in \mathcal{C}_r, \quad (5)$$

and the K frames with highest $\rho_{r,t}$ within $\mathcal{C}_r$ are retained.

Spectral token extraction. The same FFT processing yields a compact robot-level descriptor that is later injected into the fusion stage. After frame sampling, each robot supplies K tunable visual tokens $\{\mathbf{h}_{r,k}\}_{k=1}^K$ produced by MLLM visual encoder. We compute the magnitude spectrum of this short sequence, retain its low-frequency coefficients, and project them through a small MLP with LayerNorm to obtain $\mathbf{F}_r \in \mathbb{R}^{d_F}$. $\mathbf{F}_r$ encodes how each robot’s evidence evolves over the sampled subset, and is consumed by the fusion module as a per-robot temporal summary.

Spectral and Physics-Informed Multi-Robot Fusion (SPI-MRF). Single-robot video understanding mainly relies on temporal context within one coherent viewpoint. In contrast, multi-robot reasoning requires integrating geometrically constrained and mutually dependent observations. Robots may

view the same object from different angles, suffer different occlusions, or share overlapping fields of view, so the model must reason about co-visibility, relative pose, and observation reliability.

Naive fusion methods such as concatenation, addition, or self-attention over pooled tokens treat robot observations as independent and interchangeable. They therefore cannot reliably resolve viewpoint conflicts, infer shared scene structure, or update team-level beliefs when observations disagree. SPI-MRF is designed to explicitly model these cooperative dependencies.

Given the sampled visual tokens $\{\mathbf{h}_{r,k}\}_{k=1}^K$ from each robot through probabilistic head, we first obtain a robot-level visual state (μ_r, σ_r) , where μ_r encodes visual evidence and σ_r estimates variance to quantify the reliability. To capture temporal dynamics, we compute a Fourier embedding $\mathbf{F}_r$ from the sampled tokens, and encode robot motion commands into a pose embedding $\mathbf{P}_r$. We then form a physics-aware robot token by combining pose, spectral, and robot-role embeddings as Eq. 6, using linear projection layers $\mathbf{W}_p$ and $\mathbf{W}_f$.

$$\mathbf{A}_r = \mathbf{W}_p \mathbf{P}_r + \mathbf{W}_f \mathbf{F}_r + \mathbf{E}_r. \quad (6)$$

The visual states attend to these auxiliary tokens through cross-attention [9] according to Eq. 7.

$$\hat{\mu}_{1:N} = \text{CrossAttention}(\mu_{1:N}, \mathbf{A}_{1:N}, \mathbf{A}_{1:N}), \quad (7)$$

allowing each robot’s view to be updated with motion, identity, and temporal cues. Finally, the fused robot states are aggregated into a global team belief using reliability-aware pooling and linear projection layer $\mathbf{W}_b$ as in Eq. 8.

$$\mathbf{z} = \mathbf{W}_b \sum_{r=1}^N \frac{\exp(-\|\sigma_r\|_2)}{\sum_j \exp(-\|\sigma_j\|_2)} \hat{\mu}_r. \quad (8)$$

The resulting representation $\mathbf{z}$ is used as prefix context for the language decoder, enabling cooperative question answering over aligned multi-robot observations.

Physics-Aligned Prompt-Space Distillation (PAPSD).

We propose PAPSD, a privileged-information learning framework that uses simulator-provided ground-truth robot poses during training but requires no pose input at inference. We construct physics-based teacher prompts from robot poses to encode each robot’s state and inter-robot spatial configuration. Learnable student prompts are then trained to align with these teacher prompts, distilling pose-induced physical awareness into task-adaptive latent representations. In inference, only the student prompts are used, enabling physics-aware multi-robot fusion from egocentric videos alone. The fused vector $\mathbf{z}$ conditions the language backbone LM_ϕ via projected prefix tokens, which are physically aligned through distillation from a trajectory-aware teacher used only at training time. Both teacher and student output L prompt vectors of dimension d , add learned positional codes, and apply LayerNorm. The teacher uses privileged inputs, *i.e.*, ground-truth pose embeddings $\bar{\mathbf{p}}^{\text{emb}}$ and learnable prefix tokens $\bar{\mathbf{C}}$ of the teacher, to emit a physically informed prompt as Eq. 9.

$$\mathbf{P}_T = \mathbf{g}_T(\bar{\mathbf{p}}^{\text{emb}}, \bar{\mathbf{C}}) \in \mathbb{R}^{L \times d}. \quad (9)$$

The student counterpart has access only to test-time quantities, *i.e.*, the global belief $\mathbf{z}$ and emits $\mathbf{P}_S = \mathbf{g}_S(\mathbf{z}, \mathbf{C})$, where $\mathbf{C}$ are learnable student prompts. The $\mathbf{P}_S$ and $\mathbf{P}_T$ are aligned with a distillation loss $\mathcal{L}_{\text{distill}}$ formulated by cosine similarity distance. In inference, the teacher is discarded; the student alone produces the prompt that conditions the language model, as shown in Eq. 10.

$$\tilde{\mathbf{H}} = [\mathbf{W}_z \mathbf{z}; \mathbf{W}_P \mathbf{P}_S; \text{Embed}(\hat{\mathbf{q}})], \quad \mathbf{y} = \text{LM}_\phi(\tilde{\mathbf{H}}), \quad (10)$$

where $\mathbf{W}_z$ and $\mathbf{W}_P$ are linear projection layers. Embed indicates the tokenizer and $\hat{\mathbf{q}}$ denotes original query. Because distillation happens in prompt space rather than output space, the student is supervised to recover the physical latent structure that would have best conditioned the LM, not merely its final answer – this proves markedly more sample-efficient than standard response distillation.

Training objective. We train end-to-end with a composite objective:

$$\mathcal{L} = \lambda_{\text{LM}} \mathcal{L}_{\text{LM}} + \lambda_d \mathcal{L}_{\text{distill}}, \quad (11)$$

where $\mathcal{L}_{\text{LM}}$ is the next-token cross-entropy on $\mathbf{y}$ and $\mathcal{L}_{\text{distill}}$ is the physics-informed prompt distillation loss. The loss weights λ_{LM} and λ_{distill} are selected using grid search.

Table 2: CoopSR benchmark results. We report accuracy for four reasoning tiers (T1–T4), different team sizes ($N = 2, 3, 4$), and the overall average. Note: AVG is computed over all QA samples, rather than by averaging across task types or robot-team sizes.

Datasets Models	T1	T2	T3	Habitat [48]				iGibson [32]			
	T1	T2	T3	T4	N=2	N=3	N=4	T1	T2	T3	T4
Zero-shot evaluations											
Qwen2.5-VL-7B-Instruct [2]	23.57	39.61	28.40	34.35	34.66	34.99	31.03	33.51	39.44	50.89	28.12
Qwen2.5-VL-72B-Instruct [2]	18.84	43.57	19.82	33.61	30.91	33.18	31.15	31.75	18.84	43.57	19.82
LLaVA-NeXT-Video-7B [72]	26.76	36.35	33.30	27.09	31.82	31.49	30.57	31.27	21.11	30.95	32.37
InternVL3-8B [59]	20.77	49.86	36.08	43.01	41.03	42.04	41.57	41.56	28.15	52.25	34.80
VideoLLaMA3-7B [68]	23.96	44.59	37.86	44.03	40.84	42.16	40.52	41.17	19.44	46.28	39.21
VideoLLaMA3-2B [68]	21.26	41.69	35.10	37.83	33.77	38.02	39.08	37.04	12.78	55.51	31.61
Qwen3-VL-8B-Instruct [1]	20.77	45.65	37.32	39.25	42.18	39.35	36.46	39.24	35.56	47.72	37.66
SFT baselines											
Qwen2.5-VL-7B [2]	46.67	76.30	53.02	72.53	75.04	63.82	61.77	66.68	80.00	65.77	54.10
Qwen2.5-VL-72B [2]	43.96	74.47	49.90	68.66	71.05	60.88	59.56	63.65	73.89	70.39	49.09
LLaVA-NeXT-Video-7B [72]	20.97	55.43	47.35	49.25	56.19	50.91	48.42	51.72	18.33	53.27	47.57
InternVL3-8B [59]	39.03	75.51	47.60	68.97	69.86	61.76	58.25	63.12	79.44	69.35	47.60
VideoLLaMA3-7B [68]	19.52	62.58	42.91	63.22	61.66	53.06	49.85	54.68	13.89	55.95	47.72
VideoLLaMA3-2B [68]	20.68	59.42	42.63	63.60	61.09	52.40	48.92	53.95	15.56	53.12	44.68
Qwen3-VL-8B-Instruct [1]	30.53	67.83	46.83	67.55	65.95	57.90	55.49	59.62	59.44	57.44	48.33
RAG Baselines (Best SFT model adopted)											
VideoRAG [26] (Qwen2.5-VL-7B-Instruct [2])	39.13	76.52	51.08	71.37	71.34	63.19	61.68	65.25	67.78	70.68	51.82
VideoRAG [26] (InternVL3-8B [59])	22.03	72.92	45.46	67.00	66.69	57.92	55.53	59.88	12.22	64.14	45.90
EgoRAG [63] (Qwen2.5-VL-7B-Instruct [2])	48.31	74.93	51.80	71.36	73.51	63.61	60.37	65.63	77.78	68.15	54.41
EgoRAG [63] (InternVL3-8B [59])	39.90	74.71	47.40	68.47	69.77	60.67	58.20	62.70	72.22	69.05	44.98
Keyframe Baselines (Best SFT model adopted)											
TSPO [57] (Qwen2.5-VL-7B-Instruct [2])	37.97	75.80	46.68	68.55	69.28	61.44	57.94	62.72	86.11	67.56	45.29
TSPO [57] (InternVL3-8B [59])	42.51	76.21	50.77	70.79	72.21	63.25	60.35	65.09	79.44	72.02	49.54
KTV [56] (Qwen2.5-VL-7B-Instruct [2])	19.71	52.32	41.68	57.11	50.56	49.63	47.33	49.13	21.67	53.12	43.47
KTV [56] (InternVL3-8B [59])	19.13	61.43	40.26	62.95	60.60	51.92	48.61	53.53	13.89	58.63	40.88
Our proposed approach											
SP-CoR (InternVL3-8B [59])	57.78	78.31	56.98	74.96	77.07	68.07	65.37	69.99	76.11	73.81	55.17
SP-CoR (Qwen3-VL-8B-Instruct [1])	57.20	78.14	58.38	74.54	78.28	67.32	65.31	70.11	79.44	72.77	59.42
SP-CoR (Qwen2.5-VL-7B-Instruct [2])	55.36	78.72	59.36	74.94	78.26	67.34	66.60	70.55	74.44	71.88	58.36

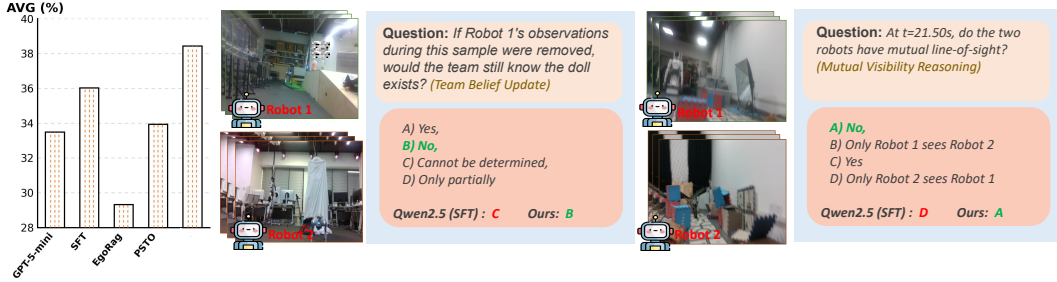


Figure 3: Performances of MLLMs on the real-world test set. Histograms on the left show the accuracy of different MLLMs. Samples on the right are qualitative samples.

5 Experimental Results

Implementation details. All experiments are conducted on 4 NVIDIA A100 GPUs. We precompute CLIP and Qwen visual features before training. The model is trained for 10 epochs using AdamW [40] with learning rate 1×10^{-4} , per-GPU batch size 16, gradient accumulation 4, and gradient clipping 1.0. Learnable prompts number is 8, M , w and K are selected as 32, 4 and 8, LoRA fine-tuning on MLLM decoder is applied with rank 8, $\alpha = 16$, and dropout 0.05. λ_{LM} and $\lambda_{distill}$ are selected as 1.0 and 0.3. Hyperparameters are selected through grid searching and validation set performance. The number of trainable parameters of SP-CoR with Qwen2.5-VL-7B [24] is 23.8M.

Analysis of the benchmark. Table 2 compares different model groups on CoopSR across Habitat [48] and iGibson [32], covering four reasoning tiers (T1–T4) and team sizes $N \in \{2, 3, 4\}$. Zero-shot MLLMs obtain low average accuracies, mostly between 31% and 42%, showing that general-purpose MLLMs struggle with cooperative spatial reasoning from multi-robot egocentric videos. SFT substantially improves performance, for example, Qwen2.5-VL-7B-Instruct [2] increases from 33.51% to 66.68% on Habitat and from 33.95% to 63.67% on iGibson [32]. However, its performance still drops for larger teams, indicating limited generalizability to increased viewpoint redundancy and cross-robot inconsistency. RAG and keyframe baselines such as EgoRAG [63] and TSPO [57] further improve evidence selection, but they mainly operate at the input level and do not explicitly model robot identity, inter-robot geometry, or cooperative view fusion. In contrast, SP-CoR achieves the best overall performance across both environments. The strongest SP-CoR variant reaches 70.55% on Habitat and 70.82% on iGibson, outperforming the strongest non-SP-CoR baselines by 3.87% and 7.12%, respectively. These gains indicate that the key challenge in CoopSR is not only selecting relevant frames, but also forming a consistent multi-robot scene representation. SP-CoR

Table 3: Ablation results on the Qwen2.5 backbone [2] for Habitat and iGibson test sets.

Method	Habitat [48]				iGibson [32]			
	N=2	N=3	N=4	AVG	N=2	N=3	N=4	AVG
w/o Spectral Energy Ranking in SE-MR ² FS	77.63	65.52	64.55	69.03	67.80	69.73	67.85	68.44
w/o Fourier Embed. in SPI-MRF	73.95	64.65	62.57	66.88	59.93	61.39	60.03	60.43
w/o SE-MR ² FS	72.53	63.75	61.00	65.58	61.94	65.98	59.51	62.40
w/o SPI-MRF	75.23	64.87	62.22	67.24	51.42	59.04	55.00	55.12
w/o PAPSD	73.89	63.69	61.70	66.24	60.57	63.26	61.60	61.79
Ours	78.26	67.34	66.60	70.55	72.74	69.92	69.85	70.82

Table 4: Cross-team-size generalizability analysis on Habitat and iGibson when we train the model on $N = 2$ and $N = 3$ while we test on $N = 4$. Note that the averaged ACC is calculated on the averaged performance of all QAs.

Method	Habitat [48]					iGibson [32]				
	T=1	T=2	T=3	T=4	AVG	T=1	T=2	T=3	T=4	AVG
SFT (Qwen2.5-VL-7B-Instruct [2])	18.84	62.39	41.81	58.24	52.30	15.00	63.44	37.08	48.24	47.18
SFT (Qwen3-VL-8B-Instruct [1])	22.03	60.65	41.30	58.68	52.09	41.67	57.71	45.42	53.85	52.22
SFT (InternVL3-8B [59])	29.86	69.49	47.32	62.81	58.23	45.00	66.96	42.92	56.09	54.91
EgoRAG [63] (Qwen2.5-VL-7B-Instruct [2])	24.06	48.41	37.54	54.12	46.12	53.33	54.19	39.17	45.35	46.22
EgoRAG [63] (Qwen3-VL-8B-Instruct [1])	22.32	47.17	39.86	53.97	46.24	21.67	38.77	47.50	39.90	40.31
EgoRAG [63] (InternVL3-8B [59])	19.13	53.91	39.35	56.38	48.65	26.67	59.47	35.42	43.91	44.31
TSPO [57] (Qwen2.5-VL-7B-Instruct [2])	23.33	63.19	38.19	62.83	53.63	23.33	45.00	45.00	58.64	51.35
TSPO [57] (Qwen3-VL-8B-Instruct [1])	20.87	51.81	40.58	56.82	48.71	11.67	57.27	44.17	43.75	44.83
TSPO [57] (InternVL3-8B [59])	20.58	56.96	39.71	56.72	49.80	18.33	59.47	43.33	52.24	50.04
SP-CoR (Qwen2.5-VL-7B-Instruct [2])	50.14	70.94	51.81	68.79	63.56	71.67	66.96	51.25	71.15	66.20
SP-CoR (Qwen3-VL-8B-Instruct [1])	48.12	71.59	54.57	68.20	64.11	66.67	68.72	56.67	70.99	67.33
SP-CoR (InternVL3-8B [59])	51.59	72.39	53.48	69.53	64.79	81.67	70.48	55.42	73.08	69.33

addresses this with three complementary biases: spectral sampling captures informative temporal changes for action and visibility reasoning; physics-informed fusion aligns observations across changing viewpoints; and prompt distillation transfers trajectory-level geometric supervision to the student model without requiring ground-truth poses at test time. Together, these components improve cross-view consistency and team-level reasoning, leading to stronger gains on T4.

Analysis of ablation study. Table 3 ablates each component of **SP-CoR** with the Qwen2.5-VL-7B-Instruct backbone [2]. The full model achieves the best average accuracy on both Habitat [48] and iGibson [32], with 70.55% and 70.82%, respectively. Removing any component consistently degrades performance, showing that spectral sampling, Fourier temporal encoding, spectral-aware fusion, and physics-informed prompt distillation are all beneficial. The largest drop occurs on iGibson [32] when SPI-MRF is removed, reducing accuracy from 70.82% to 55.12%, highlighting the importance of robust cross-view fusion in complex interactive scenes. Overall, the ablation confirms that these components are complementary and jointly improve multi-robot cooperative spatial reasoning.

Cross-team-size generalization. Table 4 evaluates generalization from smaller teams ($N = [2, 3]$) to an unseen larger team size ($N = 4$). This setting is challenging due to increased observation redundancy, viewpoint inconsistency, and more complex inter-agent relations. Across Habitat [48] and iGibson [32], **SP-CoR** achieves the best performance for all three backbones. The strongest **SP-CoR** variant reaches 64.79% on Habitat, improving over the best baseline by 6.56%, and 69.33% on iGibson [32], with a larger gain of 14.42%. These results show that **SP-CoR** generalizes more effectively than SFT, EgoRAG [63], and TSPO [57]. Beyond task adaptation, retrieval, or keyframe selection, **SP-CoR** combines dynamic spectral sampling with physics-aware fusion and prompt distillation, explicitly modeling robot relations, co-visibility, and cross-view alignment. These cooperative inductive biases enable stronger transfer to the unseen $N = 4$ setting.

Real-world generalizability. Figure 3 shows that **SP-CoR** achieves the best real-world accuracy, outperforming GPT-5 mini [55], SFT, EgoRAG [63], and TSPO [57]. Qualitative examples further show that **SP-CoR** handles team belief updates and mutual visibility reasoning more reliably than SFT, indicating stronger generalizability to viewpoint changes, partial observations, and multi-robot spatial relations in real-world situations.

6 Conclusion

We introduced **CoopSR**, a benchmark for multi-robot cooperative egocentric spatial reasoning, requiring models to integrate synchronized views for spatial, temporal, visibility, and coordination reasoning. We further proposed **SP-CoR**, which combines spectral frame sampling, physics-aware

fusion, and prompt distillation to build coherent team-level representations without requiring privileged pose information at test time. Experiments on Habitat, iGibson, cross-team-size evaluation, and real-world robot data show that **SP-CoR** consistently outperforms strong MLLM, SFT, RAG, and keyframe baselines, highlighting the importance of physically grounded cooperative reasoning for embodied AI.

Acknowledgment

The project is funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) – SFB 1574 – 471687386. This work was performed on the HoreKa supercomputer funded by the Ministry of Science, Research and the Arts Baden-Württemberg and by the Federal Ministry of Education and Research. The authors also acknowledge support by the state of Baden-Württemberg through bwHPC and the German Research Foundation (DFG) through grant INST 35/1597-1 FUGG. This project is also supported in part by the National Natural Science Foundation of China under Grant No. 62473139, in part by the Hunan Provincial Research and Development Project (Grant No. 2025QK3019), and in part by the State Key Laboratory of Autonomous Intelligent Unmanned Systems (the opening project number ZZKF2025-2-10). This research was partially funded by the Ministry of Education and Science of Bulgaria (support for INSAIT, part of the Bulgarian National Roadmap for Research Infrastructure).

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A Technical Appendices and Supplementary Material

A.1 Society Impact and Limitations

This work advances cooperative spatial reasoning for multi-robot teams that jointly interpret synchronized egocentric observations. By enabling robots to integrate partial viewpoints into a shared scene understanding, **CoopSR** may support important applications in search and rescue, assistive robotics, smart homes, warehouse automation, infrastructure inspection, and navigation in hazardous or unknown environments. In these settings, stronger team-level situational awareness can help robots cover larger areas, resolve occlusions, coordinate tasks more effectively, and reduce human exposure to risk.

At the same time, multi-robot egocentric perception introduces privacy and safety challenges. Robots operating in homes, workplaces, hospitals, or public spaces may capture sensitive visual information, so deployment should follow data minimization, consent, anonymization, secure storage, and access-control practices. In addition, higher benchmark accuracy should not be interpreted as a guarantee of safe autonomous behavior. False spatial reasoning, incorrect visibility estimation, or wrong task assignment could lead to unsafe navigation, collisions, failed coordination, or inappropriate actions in physical environments. Therefore, real-world deployment should include human oversight, uncertainty estimation, fail-safe mechanisms, and extensive validation under diverse operating conditions.

Our dataset is primarily simulation-based, with a controlled real-world quadruped robot test set. While this enables scalable and systematic evaluation, it may not fully capture open-world variation such as clutter, lighting changes, sensor noise, social context, larger robot teams, or unseen embodiments. Cooperative robot reasoning also has dual-use potential: it can improve rescue, accessibility, and industrial safety, but could also be misused for surveillance or intrusive monitoring. Future work should therefore evaluate robustness, privacy preservation, fairness, and human-centered safety alongside accuracy.

Limitation Discussion. Although **CoopSR** covers diverse simulated environments, the real-world test set remains relatively limited in scene diversity, robot types, and team sizes. Future work should extend the benchmark to larger real-world robot teams, more open-ended tasks, and safety-critical scenarios with richer physical interactions.

A.2 Details of Simulator and Dataset Splits



Figure 4: Overview of data collection devices. We make use of two Unitree quadruped robots and two different motion capture systems from FZMotion and NOKOV. We use daily life objects to set up 13 different scenarios in these two data collection rooms, which can provide ground truth pose data for the robots.

Habitat scenes. Habitat [48] is a high-performance embodied AI simulation platform that supports photorealistic indoor navigation, egocentric RGB-D observation, and scalable agent-environment interaction. Robots are deployed in 154 procedurally generated indoor floor plans. Each scene is explored by $N \in \{2, 3, 4\}$ robots executing a coverage exploration policy. Synchronised RGB video, semantic segmentation mask for each frame, depth video, odometry commands, and ground-truth poses are recorded. The trajectory log for each exploration is stored together with the co-located video files, containing per-timestep columns: position, heading, forward velocity, and angular velocity.



Figure 5: An overview of the world clouds of the QAs from (a) Habitat, (b) iGibson, and (c) Real-world lab.

iGibson scenes. IGibson [32] is a realistic interactive simulation platform for embodied AI, supporting indoor navigation, physics-based interaction, and egocentric RGB-D perception in diverse household environments. 14 interactive scenes are adopted for the data collection. During the exploration, we also adopt $N \in \{2, 3, 4\}$ for the data collection.

Train / val / test splits. For each configuration, scenes are divided into disjoint training, validation, and test splits, with 119/12/23 scenes for Habitat settings and 8/2/4 scenes for iGibson. For the real-world quadruped robot set, all collected scenes are used for testing. No scene is shared across splits, thereby preventing scene-level data leakage.

A.3 Dataset Annotation Details, Diversity and Statistical Analysis

A.3.1 Human Checking Details

To ensure the reliability of **CoopSR**, we conduct a multi-stage human verification process. Four human annotators independently review the generated QA pairs using synchronized multi-robot egocentric videos, simulator metadata, and the proposed answer choices. Annotators verify whether each question is visually grounded, whether the answer is uniquely determined, and whether the question genuinely requires cooperative reasoning across multiple robot viewpoints rather than single-view recognition or language priors.

For each QA pair, annotators either accept it, correct it, or discard it. Corrections include resolving robot identity confusion, inaccurate spatial relations, ambiguous object references, incorrect visibility states, and temporal boundary inconsistencies. QA pairs are discarded if the target object is not sufficiently observable, if multiple answer choices are plausible, if the answer can be inferred from textual priors alone, or if a cross-robot level question does not require cross-robot evidence despite belonging to a cooperative reasoning category. To further improve benchmark quality, QA pairs marked ambiguous by at least two annotators are removed.

A.3.2 Dataset Diversity and Statistical Analysis

We analyze the linguistic and semantic diversity of **CoopSR** in Figures 5 and 6. Figure 5 shows word clouds for the Habitat [48], iGibson [32], and real-world lab subsets. Across domains, frequent terms such as “timestep”, “observations”, “robot”, “object”, “team”, and “between” indicate a shared focus on temporally grounded multi-robot egocentric reasoning. Habitat emphasizes spatial relations with terms such as “left”, “right”, “distance”, and “closest”, whereas iGibson [32] highlights interaction-rich household scenarios through words such as “chair”, “cabinet”, “table”, “action”, and “pushing”. The real-world subset contains prominent terms such as “pose”, “nearest”, “farthest”, “field”, “region”, and “depth”, reflecting visibility-, distance-, and pose-aware reasoning.

Overall, Figure 5 suggests that **CoopSR** spans diverse simulation and real-world language patterns while consistently targeting cooperative spatial, temporal, visibility, and action reasoning. Figure 5 provides quantitative statistics of the QA and object distributions. Figure 6a reports the number of QA samples for the 19 QA types. Most QA categories contain approximately $7K$ – $8K$ samples, indicating broad and relatively balanced coverage across the benchmark taxonomy. A few categories contain fewer samples because they correspond to more constrained events or higher-level cooperative reasoning cases that require specific scene configurations, such as rare interaction events, team-level belief updates, or task-assignment conditions. This distribution ensures that the benchmark

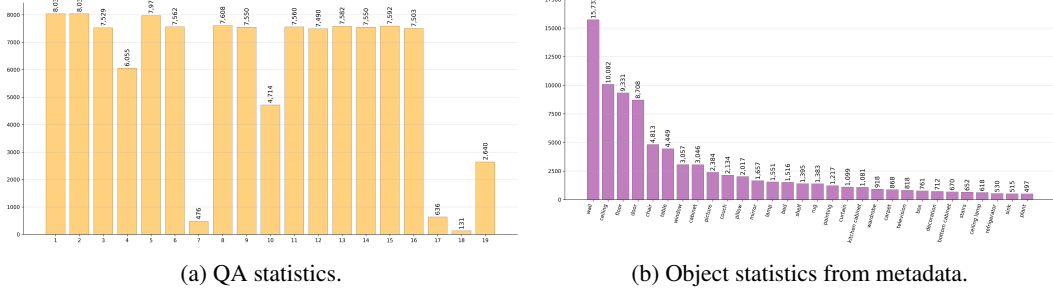


Figure 6: An overview of the statistics of (a) the number of QAs and (b) the number of objects.

Table 5: Ablation studies of SPI-MRF and PAPSD.

(a) Ablation study of the fusion strategy used in SPI-MRF.

Method	Habitat [48]				iGibson [32]			
	N=1	N=2	N=3	AVG	N=1	N=2	N=3	AVG
Concatenation	72.17	60.55	56.66	62.89	60.66	62.23	62.12	61.67
Addition	65.61	58.61	54.56	59.43	46.75	51.64	51.95	50.14
Multiplication	69.24	59.87	56.76	61.77	51.51	57.73	55.08	54.76
SelfAttention	75.65	66.80	65.00	68.99	64.78	67.48	66.46	66.23
Ours	78.26	67.34	66.60	70.55	72.74	69.92	69.85	70.82

(b) Ablation of the loss weight for L_{distill} ranging from $[0.1, 0.7]$.

L_{distill} in PAPSD	Habitat [48]				iGibson [32]			
	N=1	N=2	N=3	AVG	N=1	N=2	N=3	AVG
0.1	77.63	67.62	64.86	69.84	68.25	68.79	67.33	68.11
0.3	78.26	67.34	66.60	70.55	72.74	69.92	69.85	70.82
0.5	72.19	71.13	70.81	70.54	72.19	71.13	70.81	71.37
0.7	78.12	66.84	65.76	70.05	72.46	72.63	69.50	71.49

contains sufficient supervision for common spatial and temporal reasoning tasks while still including challenging long-tail cooperative reasoning cases.

Figure 6b summarizes the object-frequency distribution extracted from metadata. The object distribution is naturally long-tailed: structural and frequently visible categories such as walls, chairs, floors, doors, and cabinets appear most often, while smaller or scene-specific objects such as pillows, lamps, mirrors, sinks, flowers, televisions, and decorative objects occur less frequently. This long-tail distribution reflects realistic indoor environments, where common background structures dominate visual observations while many task-relevant objects appear sparsely. Such diversity is important for evaluating whether MLLMs can reason beyond frequent object categories and generalize to less common objects under partial, egocentric, and multi-view observations.

Together, Figures 5 and 6 demonstrate that **CoopSR** provides both linguistic diversity and semantic richness. The benchmark is not limited to simple object recognition or single-view spatial questions; instead, it contains diverse question formulations, multiple reasoning types, varied object categories, and both simulated and real-world domains. These statistics support the effectiveness of **CoopSR** as a comprehensive benchmark for evaluating cooperative multi-robot egocentric spatial reasoning.

A.4 More Ablation Analysis of our SP-CoR Approach on Hyperparameters

We further analyze the sensitivity of **SP-CoR** to three important hyperparameters: the distillation loss weight in PAPSD, the number of learnable prompts, and the number of dense frames retained during CLIP-based similarity pre-selection. The results are reported on both Habitat [48] and iGibson [32] across different robot-team sizes. Table 5a compares different fusion strategies in SPI-MRF. Overall, the proposed fusion strategy consistently achieves the best performance on both Habitat and iGibson, with average accuracies of 70.55% and 70.82%, respectively. Compared with simple concatenation, addition, and multiplication, our method provides more effective multi-view feature integration, indicating that directly combining view features is insufficient to fully capture complementary spatial cues. Self-attention performs better than these simple fusion baselines, suggesting that adaptive view-wise feature interaction is beneficial. However, our method further improves over self-attention by 1.56% on Habitat [48] and 4.59% on iGibson [32] in terms of average accuracy, demonstrating its stronger capability in modeling cross-view relationships and selecting informative features across different numbers of observed views.

First, we vary the weight of the prompt-space distillation loss, λ_d , from 0.1 to 0.7 in Table 5b. The results show that PAPSD is effective across a broad range of weights. On Habitat [48], the average accuracy remains stable, ranging from 69.84% to 70.55%, indicating that the model is

Table 6: Ablation studies of dense frame selection and learnable prompts.

(a) Ablation for dense frame selection during the CLIP-based similarity pre-selection. (b) Ablation of the number of learnable prompts ranging from [4, 10].

#Dense Frames	Habitat [48]				iGibson [32]			
	N=1	N=2	N=3	AVG	N=1	N=2	N=3	AVG
16	76.91	67.66	65.88	69.98	69.26	69.17	67.94	68.77
32	78.26	67.34	66.60	70.55	72.74	69.92	69.85	70.82
64	77.58	68.07	65.37	69.99	72.28	68.32	69.59	70.07

#Prompts in PAPSD	Habitat [48]				iGibson [32]			
	N=1	N=2	N=3	AVG	N=1	N=2	N=3	AVG
4	77.12	67.70	66.69	70.34	69.99	69.82	67.33	69.01
6	77.20	67.66	64.88	69.73	69.99	69.82	67.16	68.95
8	77.58	68.07	65.37	69.99	72.28	68.32	69.59	70.07
10	77.65	68.05	65.21	70.11	68.07	69.82	68.55	68.80

Table 7: Comparison to text-only baseline on CoopSR benchmark.

Datasets Models	Habitat [48]				iGibson [32]			
	N=2	N=3	N=4	AVG	N=2	N=3	N=4	AVG
Text-Only Baseline	23.51	27.27	29.97	29.47	26.58	30.31	31.30	26.97
SFT (Qwen2.5-VL-7B-Instruct [2])	75.04	63.82	61.77	66.68	63.40	63.92	63.68	63.67
SP-CoR (Qwen2.5-VL-7B-Instruct [2])	78.26	67.34	66.60	70.55	72.74	69.92	69.85	70.82

Table 8: Comparison of our approach and the counterpart under robot Dropout setting.

Datasets Models	Habitat [48]					iGibson [32]				
	T=1	T=2	T=3	T=4	AVG	T=1	T=2	T=3	T=4	AVG
Robot Dropout on SP-CoR (Qwen2.5-VL-7B-Instruct [2])	31.69	58.29	44.10	59.74	53.29	48.89	52.38	43.31	48.64	48.35
SP-CoR (Qwen2.5-VL-7B-Instruct [2])	55.36	78.72	59.36	74.94	70.55	74.44	71.88	58.36	74.63	70.82

not overly sensitive to the exact distillation weight. On iGibson [32], increasing λ_d generally improves performance, with the average accuracy increasing from 68.11% at $\lambda_d = 0.1$ to 71.49% at $\lambda_d = 0.7$. This suggests that stronger physics-guided prompt supervision is particularly beneficial in iGibson [32], where scenes contain more complex interactive dynamics. However, overly large weights may slightly reduce performance for some team-size settings, implying that the distillation objective should complement, rather than dominate, the task supervision.

Second, we study the number of learnable prompts in PAPSD as shown in Table 6b. The performance remains relatively stable when using 4 to 10 prompts, demonstrating the robustness of the prompt-space distillation design. Using 8 prompts achieves a strong balance across both environments, reaching 69.99% average accuracy on Habitat [48] and 70.07% on iGibson [32]. Increasing the number of prompts to 10 slightly improves Habitat performance to 70.11%, but decreases iGibson [32] accuracy to 68.80%. This indicates that a larger prompt capacity does not necessarily lead to better generalization, likely because excessive prompts introduce redundant latent factors or overfit to environment-specific patterns. Therefore, a moderate number of prompts provides sufficient capacity to encode distilled spatial priors while maintaining stable generalization.

Third, we evaluate the number of dense frames used during CLIP-based similarity pre-selection, as shown in Table 6a. Increasing the dense-frame budget from 16 to 32 improves the average accuracy from 69.98% to 70.55% on Habitat [48] and from 68.77% to 70.82% on iGibson [32]. This confirms that retaining more candidate frames before spectral refinement helps preserve useful temporal and semantic evidence. However, further increasing the budget to 64 does not yield additional gains, leading to 69.99% on Habitat and 70.07% on iGibson [32]. This suggests that a moderate candidate set is sufficient for capturing informative moments, while too many dense frames may introduce redundant or noisy visual evidence.

Overall, these ablations show that hyperparameter sensitivity of **SP-CoR** is acceptable. PAPSD benefits from a properly weighted distillation objective and a moderate number of learnable prompts, while the frame sampler performs best when the CLIP-based candidate set balances evidence coverage and redundancy reduction. These findings further support the design of **SP-CoR**, where physics-aligned prompt distillation and spectral frame selection jointly improve cooperative multi-robot spatial reasoning.

A.5 More Qualitative Analysis

Figure 7 compares **SP-CoR** with the SFT (Qwen2.5-VL-7B [2]) baseline on representative **CoopSR** examples. **SP-CoR** is more accurate on tasks requiring cross-robot view integration, temporal event localization, mutual visibility reasoning, and teamwise object association. The SFT baseline often confuses robot identities, action phases, or visually similar objects across views, suggesting that

standard fine-tuning mainly adapts the backbone to the QA format but does not provide explicit cooperative inductive biases.

These examples highlight the novelty of **SP-CoR**: rather than treating multi-robot videos as concatenated visual tokens, it introduces dynamics-aware frame sampling to retain informative temporal evidence, physics-informed fusion to align observations under changing egocentric viewpoints, and prompt-space distillation to transfer pose-level geometric priors without requiring poses at test time. As a result, **SP-CoR** can build a more consistent team-level scene representation, enabling robust reasoning over partial observations, inter-robot relations, and dynamic cooperative events.

A.6 Comparison with Text-Only Baseline

We include a text-only baseline to evaluate whether **CoopSR** can be solved using linguistic priors or dataset biases without visual observations. As shown in Table 7, the text-only baseline obtains much lower accuracy than vision-based models, with only 29.47% on Habitat and 26.97% on iGibson, compared with 66.68% and 63.67% for the SFT baseline. This large gap demonstrates that **CoopSR** requires grounding in synchronized multi-robot egocentric videos, while the further gains of **SP-CoR** show the benefit of explicit cooperative spatial reasoning.

A.7 Importance of Multi-Robot Cooperative Reasoning and Experiments of Missing Robot

To verify that **CoopSR** genuinely requires cooperative reasoning across multiple robots rather than relying on a single informative viewpoint, we conduct a robot-dropout analysis, where one robot stream is randomly removed during inference. Results are shown in Table 8. Removing one robot consistently degrades performance across all reasoning tiers and datasets. On Habitat, the average accuracy drops from 70.55% to 53.29%, while on iGibson [32] it decreases from 70.82% to 48.35%.

The degradation is especially clear on higher-level cooperative reasoning tasks, where models must integrate complementary observations to infer scene-level structure, mutual visibility, temporal interactions, and team belief updates. These results show that **CoopSR** cannot be reliably solved from a reduced set of robot views alone. Missing one robot removes critical evidence for resolving occlusions, cross-view object grounding, and inter-robot spatial relations. The substantial gap between Robot Dropout and full **SP-CoR** confirms that the benchmark evaluates genuine multi-robot cooperative spatial reasoning rather than independent single-robot perception.



Figure 7: Overview of qualitative results of our approach and SFT Qwen2.5-VL-7B [24].